Toan Ly
Computer Science

Daniel Lindsey
Computer Science

Advised By:
Dennis Wu
Riverain Technologies

Med Assist

An Assistive Mental Health Platform

CEAS '26 EXPO

Problem Statement

Mental health is a complex and deeply personal topic. Many people who are struggling may be hesitant to reach out for support. Cost, fear of judgment, and uncertainty all create barriers. Chatbots offer a free and stressless alternative to human interaction and create a good space to vent; but due to their propensity to hallucinate, provide misleading suggestions, and reinforce harmful thinking - they are a poor counseling tool.

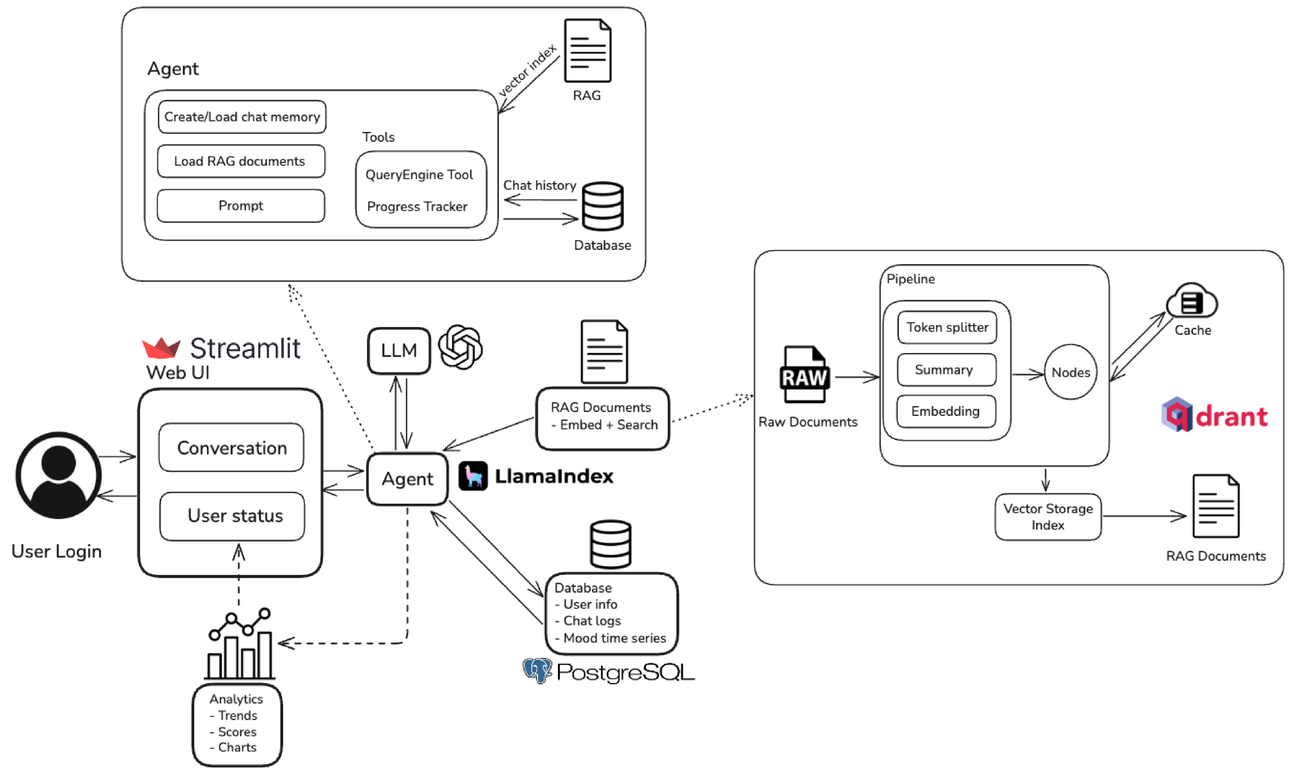
Proposed Solution

MedAssist seeks to bridge this gap and provide a middle ground between professional counseling and biased LLM chats. By focusing on reflection, emotional awareness, and grounding in Diagnostic and Statistical Manual of Mental Disorders (DSM-5) context, MedAssist provides a secure place to vent but does not suggest professional advice.

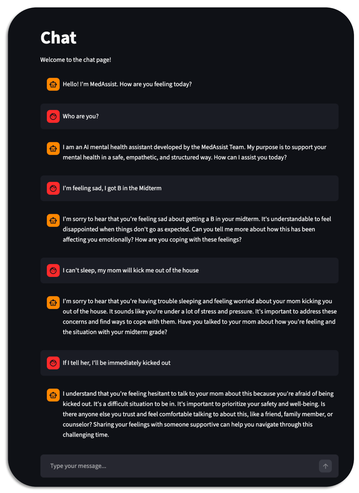
Project Abstract

MedAssist is an assistive mental health platform powered by a large language model (LLM) with retrieval-augmented generation (RAG). It offers a free, accessible, and unbiased alternative to mental health care. Our system connects an LLM to trusted DSM-5 knowledge to give accurate, context-aware responses. Through an interactive chat interface, users are able to express their thoughts, monitor mood patterns, and track personal growth over time. Alternative resources are also provided for users requiring more professional medically-focused assistance.

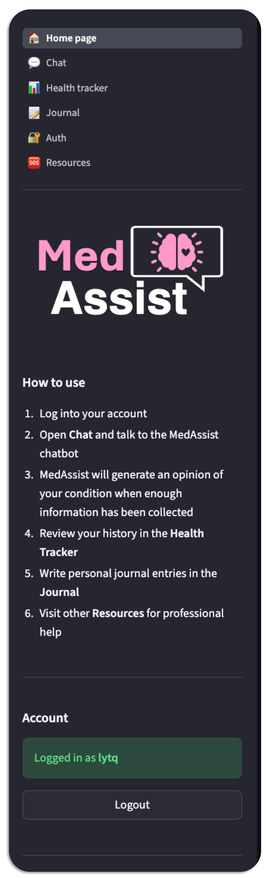
System Overview



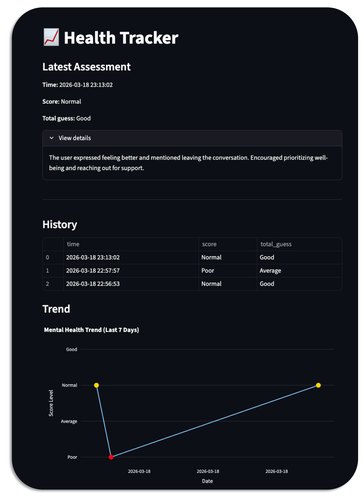
Interfaces



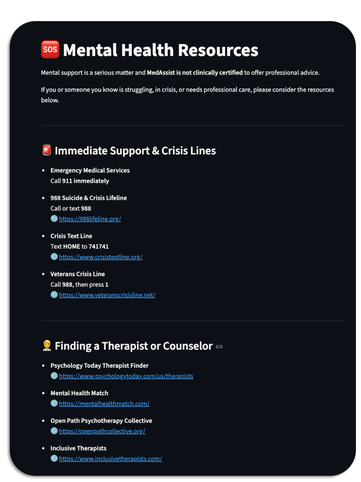
Chat Interface



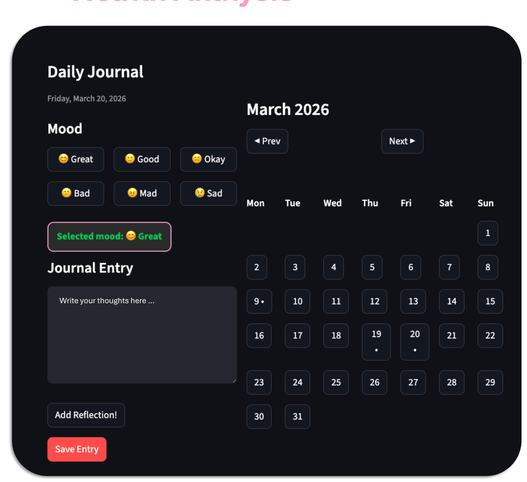
Navbar



Health Analysis



Additional Resources

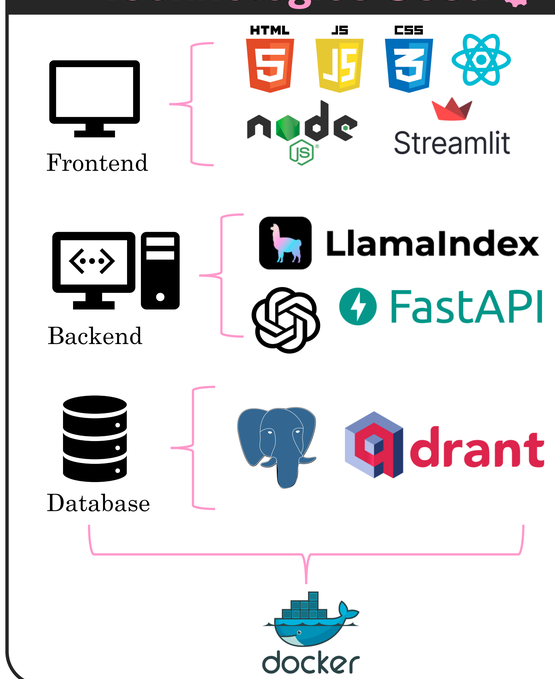


Journal & Mood Tracker

Key Features

- **DSM-5 RAG Chatbot** – Users can interact with a chatbot featuring DSM-5 context for more accurate and informative chats.
- **Journal & Mood Tracking** – Users can create daily journal entries to record their thoughts and track their moods.
- **Resources Hub** – Users can access a compiled list of third-party professional resources if they are seeking clinical or medical assistance.

Challenges

- **Data Security** – Storing minimally required information and utilizing local storage as required.
- **Chat Generation** – Reducing bias and preventing misleading or harmful chats.
- **Avoiding Misrepresentation** – Making use of clear disclaimers and proving alternative sources for professional help.

Technologies Used